SAMIA RAHMAN MRIDULA

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OBJECTIVE

Computer Science student at BRAC University with a focus on Natural Language Processing and Machine Learning, researching detecting in low-resource text using transformer models. Seeking research and professional opportunities where I can apply my skills to real-world language problems.

EDUCATION

BRAC University Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering (CSE)
Undergraduate Student **Mohammadpur Preparatory School and College** Dhaka, Bangladesh

Higher Secondary Certificate (HSC)
GPA: 5.00/5.00 **Mohammadpur Preparatory School and College** Dhaka, Bangladesh

Secondary School Certificate (SSC)
GPA: 5.00/5.00

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript
Web Development: HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, Express.js, MERN Stack, RESTful APIs
Databases: SQL, MySQL, MongoDB
Machine Learning & Data Science: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, OpenCV
AI & Deep Learning: PyTorch
Tools & Technologies: Git/GitHub.
Desgin: Figma

PROJECTS

Mass Transit Control System React.js, JavaScript, MongoDB, HTML, CSS
Academic Group Project

- Developed a full-stack web application for mass transit management, contributing to both frontend and backend development.
- Implemented lost-item management functionality and Bangla-English language switching.
- Collaborated with team members to integrate features and build an interactive user interface.
- **GitHub:** View Project

Obesity Level Prediction Using Machine Learning Python, Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib
Academic Group Project

- Analyzed and preprocessed a dataset of 2,111 records to predict seven obesity categories based on demographic, dietary, and lifestyle factors.
- Performed exploratory data analysis, handled missing values and outliers, encoded categorical features, and applied feature selection and standardization.
- Trained and compared Decision Tree, Random Forest, Logistic Regression, and Neural Network classifiers, using SMOTE to address class imbalance.
- Achieved 90.21% accuracy with the Random Forest classifier on the reported scaled test-set evaluation and visualized model performance using confusion matrices, ROC curves, and comparative metric plots.
- **GitHub:** View Project

RESEARCH EXPERIENCE

The Hidden Gamble: Online Gaming and Gambling Tendencies Among Youth Human-Computer Interaction (HCI) — Mixed-Methods Research
Undergraduate Research Project, BRAC University

- Investigated gambling-like tendencies linked to chance-based reward mechanics in online games among Bangladeshi youth.
- Conducted 30 semi-structured interviews and performed exploratory data analysis (EDA) on survey responses from 150 participants using descriptive statistics and data visualization.
- Adapted an open-source Genshin Impact Wish Simulator to prototype responsible-gaming interventions.
- Designed spending-management controls and a dynamic reward-progress indicator to promote spending awareness and transparency.
- Evaluated the prototypes with 10 participants through usability testing, focusing on usability, spending awareness, and decision-making support.
- **Project repository:** GitHub

HONORS & AWARDS

Vice Chancellor's List, BRAC University [2023]